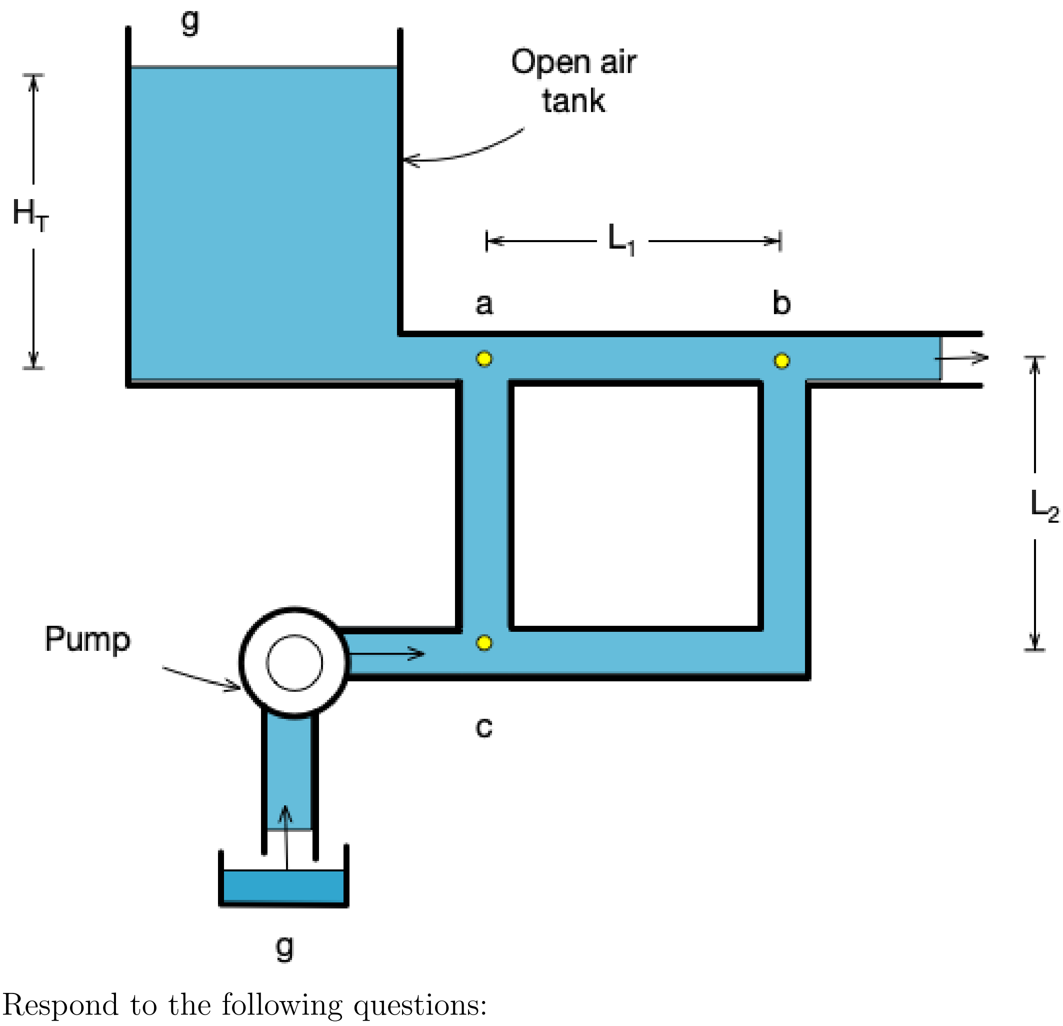
Mechatronic systems simulation [146029] - MOD2 — 2025, Jun 23^{rd}

Surname | Name | Mat. number

Signature ____________________

1 Modelling: Exercise 1

The figure shows a water distribution network which is an hydraulic system containing pipes, an overhead storage tank, a pump, and consumption at a node. We have to assume that there are no elevation differences between nodes of the system. The top of the tank is assumed to be open to the atmosphere, and considered the same as the datum or reference node g. Also the pump is connected on one side to an open air tank g. The pipes are connected between pairs of nodes a-b, c-a and c-b. We do not consider inertance components and the pipe are modelled only with resistance. They have cross section with diameter d and dynamic viscosity μ. We assume to have laminar flow.



Respond to the following questions:

1 Draw the corresponding linear graph and the normal tree.

2 write the incident matrix and derive the **circuit equations** and the **node equations**.

3 Write the list of **costitutive equations** for each element in the electrical system.

4 Assemble the equations and specify the list of input variables, the list of system states and the list of equations of dynamics.

To do the calculations you can use Maple or Matlab SymbolicToolbox. You have to upload the Maple or Matlab file.

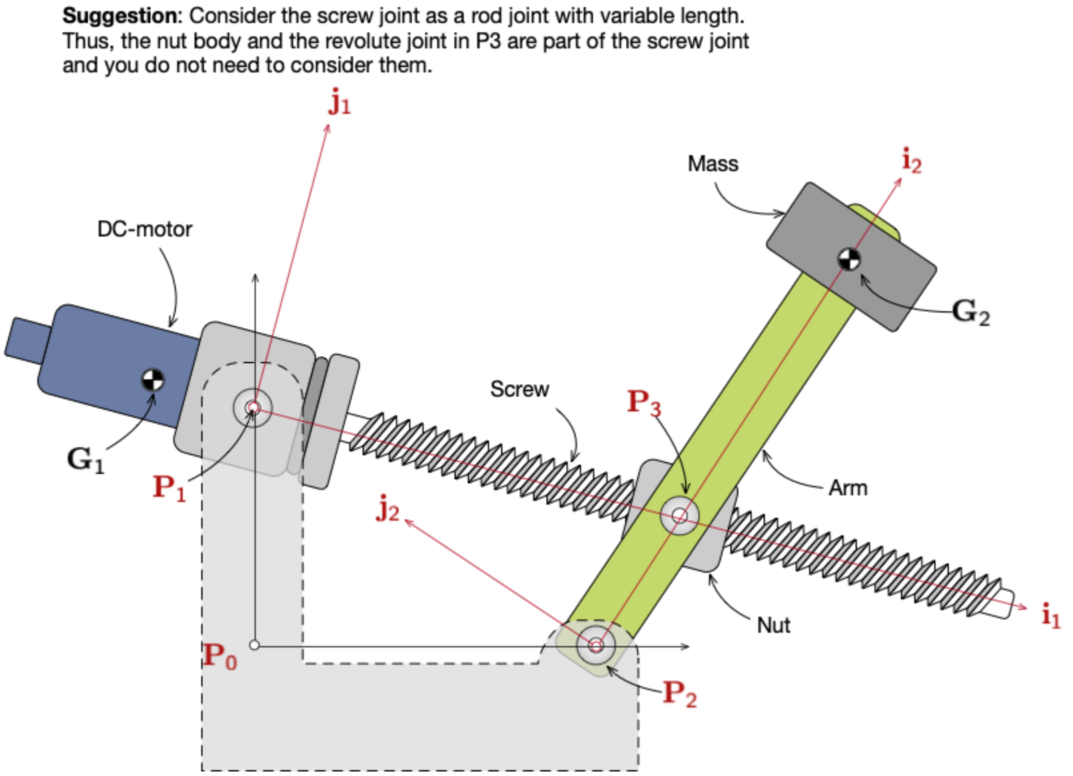
Draw the corresponding Linear Graph (LG) and the normal tree.:

Report the incidence matrix.

Report the final set of equations.

2 Modelling: Exercise 2

The figure shows the Maxpid robot subsystem which is used to gather fruit and sort of waste. It is servo controlled and can angularly orient its arm. A spindle nut type movement trasnfomation system (that uses a screw) is used to change the rotational motion of the output axis of the motor screw in the arm rotational movement. The trsnasmission ratipon of the screw is $\tau_s[mmm/rev]$. The arm is considered a single body with the mass. Thus the centre of mass and the inertia characteristics of the arm+mass element are concentrated in $\mathbf{G}_2$. The mass is m_2 and the moment of inertia I_{z_2}. The servo motor has the centre of mass in $\mathbf{G}_1$ and the mass is m_1 and the moment of inertia I_{z_1}. The positions of mechanical joints and centre of masses are given in the body reference frames indicated in figure. Specifically, point $\mathbf{G}_2 = [-x_{G1}, 0, 0]^T$, point $\mathbf{P}_3 = [L_3, 0, 0]^T$, point $\mathbf{G}_2 = [x_{G2}, 0, 0]^T$. Points $\mathbf{P}_1$ and $\mathbf{P}_2$ are the locations of the revolute joints that connect the servo-motor and the arm to the ground. Their coordinates in the ground are $\mathbf{P}_1 = [0, H_1, 0]^T$ and $\mathbf{P}_2 = [L_2, 0, 0]^T$. Gravity acts along negative y-axis.
The servo motor imposes the desired angular position with its torque T_a.



Respond to the following questions:

1 Draw the LG and the normal tree of the system

2 derive the kinematic constraint equations using the recursive formulation (i.e. loop equations)

3 derive the expression of the point G_1, G_2 and P_3 to be later used for equations of motion.

4 derive the equations of motion using the Neutown Euler approach

5 *Optional* include the screw friction in the model.

Suggestion: consider the screw and nut as a single joint that acts as a rod joint with variable length. Define the reaction forcees accordingly.
To do the calculations you can use Maple or Matlab SymbolicToolbox. You have to upload the Maple or Matlab file.

Draw the corresponding Linear Graph (LG) and the normal tree.:

Report the loop equations.

Report the final set of equations of motion.